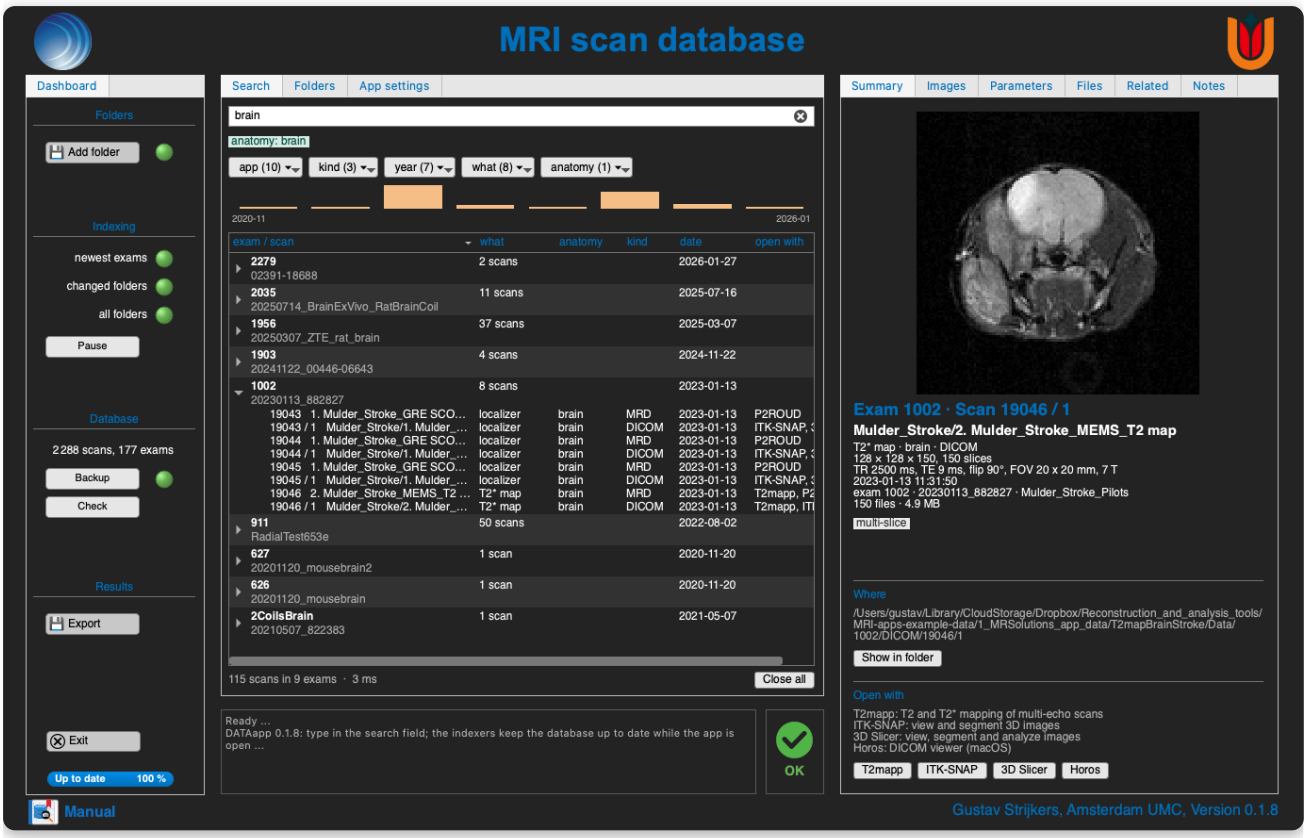


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1. Overview

DATAapp is an application with a graphical user interface that makes the MR scans you made searchable: the scans in the MR Solutions scanner's MRD and Data folders, on a network drive they are copied to, and in folders of your own. It finds every scan, reads what its files say (the protocol, the sequence, the timing, the size, the date, the subject, the project), works out what kind of scan it is and what it shows, and keeps it all in a database, so that a search answers at once however many scans there are.

For each scan DATAapp shows its images (when it has them), every value read from its files, the scans that belong with it, and the apps that can open or analyze it: Retrospective, P2ROUD, MAPapp, T1mapp, T2mapp, ADCmapp, DSCmapp, SHIMapp and TRAJapp, and programs such as ITK-SNAP, Horos and 3D Slicer. One click opens the scan in the app chosen.

DATAapp reads MR Solutions scans (MRD, SUR and DICOM files with their protocol and reconstruction files), NIfTI files, DICOM series of other preclinical scanners, pictures, HDF5 and MATLAB files, and the records, results and data files the MRI apps write. It shows preclinical MRI only: scans of people (from a clinical scanner, or of a subject over 5 kg), other modalities such as ultrasound, screenshots and photos stay out of the results. Pictures show when they are in a scanner's exam or an app made them (Retrospective's movies, the maps' pictures); HDF5, MATLAB and result files also when they lie near MRI data, in a folder whose own folder, parent folder or neighbouring folders hold scans. An HDF5 data file shows its images in **Images**.

DATAapp keeps its database up to date by itself while it is open, quietly in the background, and several people on several computers can use it at the same time.

DATAapp is a stand-alone application: everything it uses is inside it, and nothing else has to be installed.

2. Contact

DATAapp and this manual are written by:

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3. Disclaimer

This is a beta version, the first of a new application. However, this does not warrant that the functions contained in the program will meet your requirements or that the operation of the program will be uninterrupted or error-free. What DATAapp works out about a scan (what kind of scan it is, its anatomy, the apps that suit it) is its best guess from the files; check it before you rely on it, and correct it where it is wrong (section 10). DATAapp never changes, moves or removes your scans: it only reads them. In case of questions or issues, please contact Gustav Strijkers.

4. Installation

macOS

Requirements: a Mac with Apple silicon (M1 or later), macOS 26 or later.

DATAapp comes as a disk image, `DATAapp-0.2.1-macOS-arm64.dmg`. Nothing else has to be installed.

1. Open the disk image and drag **DATAapp** onto the Applications folder in its window.
2. Open it from Applications. The app is not signed with an Apple Developer ID, so the first time macOS refuses it ("Apple could not verify ..."); click **Done**. Then open **System Settings > Privacy & Security**, scroll down to the line saying DATAapp was blocked, click **Open Anyway** and confirm. From then on it opens as any other app.

   Instead of step 2, the same in Terminal, once:

   ```
   xattr -dr com.apple.quarantine /Applications/DATAapp.app
   ```

Windows

Requirements: a PC with Windows 10 or 11, 64-bit.

DATAapp comes as an installer, `DATAapp-0.2.1-setup.exe`. Run it; when Windows stops it ("Windows protected your PC"), click **More info** and then **Run anyway**. To remove the app, use **Uninstall DATAapp** in the Start menu, or **Settings > Apps**.

5. Running the software

Start DATAapp as any other application. The DATAapp logo appears while the app loads, and the window follows within seconds. The window has three columns:

- **Dashboard** (left): the folders searched, the indexing and its lamps, the database, the results and **Exit**.
- **Search** (middle): the search field, the results and, in other tabs, the folders and the settings.
- **The scan chosen** (right): its summary, images, values, files, related scans and notes.

The message box under the search says what DATAapp did; the newest message is at the top. The **STATUS** picture shows whether the app is ready, busy, or has a warning or an error. The **Manual** button at the bottom left opens this manual.

The first time DATAapp has no folders to search. On the scanner's computer it finds the scanner's MRD and Data folders by itself and starts with them. On any other computer, add the folders with **Add folder** (section 6): the network drive the scans are copied to, or a folder of your own. DATAapp then goes through them in the background; the first scans show in the results within seconds, and the rest follow while you work.

DATAapp searches as you type, and the results stay up to date while it finds new scans. Your search, the theme and the widths of the result columns are kept for the next start.

6. The folders DATAapp searches

Add folder (Dashboard, or the **Folders** tab) adds a folder to search:

- **The scanner's folders.** The MR Solutions scanner keeps its scans in two folders side by side, MRD (the raw data) and Data (the images). Choose either of them, or the folder that holds both: DATAapp takes the pair, and joins the raw data and the images of each scan. Only MRD and Data are searched: other folders and files beside them are left out, and the shared index is kept in the Data folder.
- **A network drive** the scans are copied to, or any folder of your own that does not hold both MRD and Data. DATAapp searches all its folders and finds the scans wherever they are, also when the folder holds many MRD and Data pairs deeper down and other data mixed together.



The **Folders** tab lists the folders added: whether each is **on** or **off**, its kind (**scanner** for an MRD and Data pair, **user** for others), how many scans DATAapp found in it, when it last went through all of it, the folder itself and where its shared index is. A long path is shortened in the middle; rest the pointer on it to read it whole. **Switch off** (or **Switch on**) switches the folder chosen; double-clicking a folder does the same. The search shows the scans of the folders that are on, and only those folders are indexed. A folder that is off keeps its scans in the database, so switching it on again shows them at once. A folder added is on. **Remove** takes the folder chosen off the list. **Index all now** goes through every folder of it at once, instead of at the slow pace of the background indexing.

Folders added by themselves. On the scanner's own Windows computer, DATAapp adds the scanner's MRD and Data folders the first time it starts: it reads where the scanner software keeps them, and otherwise looks for `\smis\dev` holding MRD and Data on the computer's drives (`C:\smis\dev`, `D:\smis\dev`, ...). The settings file also lists data locations (`dataLocations`, below), for example the

lab's network share: each that exists on the computer is added once, the first time DATAapp finds it. A location that does not exist there is skipped without a word, and a location you **Remove** is not added again.

One folder, several paths. The same folder can be reached by several paths, for example a network share connected under another name (`/Volumes/Gustav-1`) or through a share above it (`/Volumes/BMEPH/Gustav`). DATAapp recognizes it by its shared index and lists, searches and indexes it once: adding it again by another path adds nothing. DATAapp remembers every path the folder was reached by: when one is not connected, it uses another that is, with all its scans, so that different mounts at different moments work alike. A folder that was copied together with its shared index (by hand, to another drive) is another folder: it gets an index of its own.

Copies. Two folders that are on may hold the same scans, for example the scanner's folders and the network drive they are copied to. The search shows such a scan once, from the folder added first; the scan's **Where** names the other folders that hold it. Several smaller folders can be on together without any scan showing twice.

The shared index. Several people may search the same folder from several computers at the same time. Whatever one computer learns about the scans of a folder it writes into that folder's shared index, a folder named `.DATAapp-index`: in the Data folder of a scanner's pair, at the top of any other folder. The other computers take it from there, so that a scan is read only once for everyone, and a computer that starts DATAapp for the first time has the whole folder at once. The `.DATAapp-index` folder is hidden, on the Mac and on Windows, also on a network drive, so you don't see it while browsing your data (in the Finder, Cmd-Shift-. shows hidden folders; in the Explorer, View > Show > Hidden items). Leave it where it is. When you cannot write in a folder (a read-only network share), DATAapp searches it for you alone, without a shared index.

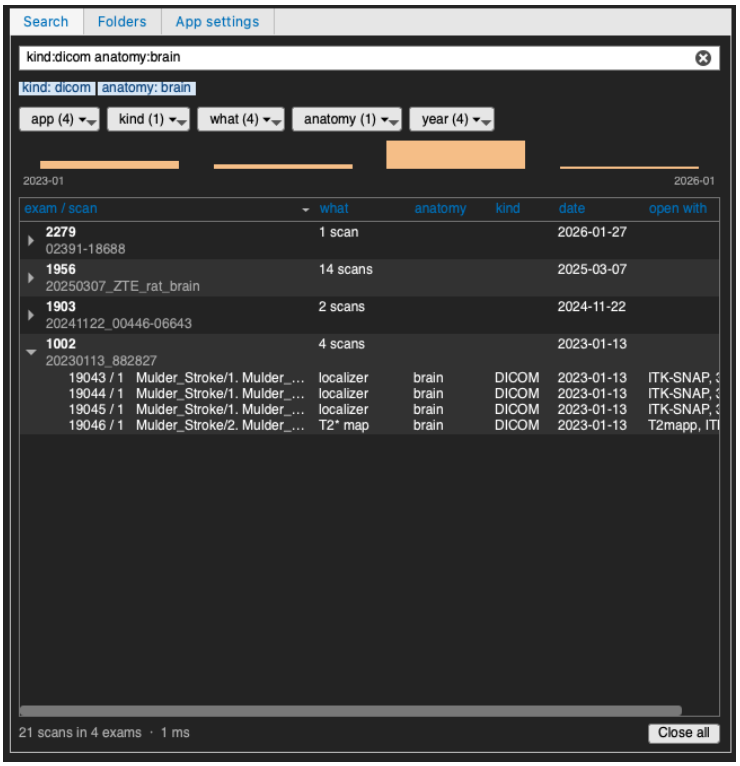
7. Searching

Type in the search field. DATAapp searches as each word is finished, and when you press Enter.

- **Words** find the scans that have them anywhere: in the protocol's name, the sequence, the exam's name, the subject, the project, the folder names, or any value read from the files. `heart cine` finds the cine scans of the heart. Words of the same meaning count too: `cardiac` finds the heart scans.
- **Parts of names.** Three letters or more find any part of a name, wherever it is in the name: a part of an exam's name, of an animal's code or of a date in a name.
- **Numbers.** An exam's number or a scan's number brings that exam to the top of the results.
- **Filters** narrow the search, written as `name:value`:

Filter	Finds
`kind:`	the kind of data: `mrd` (raw data), `dicom`, `sur`, `nifti`, `image`, `hdf5`, `matlab`, `record`, `result`
`type:`	what the scan is: `cine`, `T1 map`, `T2 map`, `diffusion`, `perfusion`, `localizer`, ... (in quotes when it holds a space: `type:"T1 map"`)
`anatomy:`	what it shows: `heart`, `brain`, `kidney`, `phantom`, ...
`app:`	the scans an app can open: `app:T1mapp`
`exam:`, `scan:`	an exam's number or name, a scan's number
`date:`, `after:`, `before:`	`date:2025-04` (a month), `after:2025-01-01`, `before:2026`
`user:`, `project:`, `species:`	the exam's user, project, species
`sequence:`, `protocol:`, `path:`	a part of the sequence's or protocol's name, or of the folder
`tag:`	a tag shown with the scan: `tag:navigator`, `tag:3D`

Under the search field the words DATAapp understood are shown as coloured chips: a filter, a kind of scan, an anatomy, an app, or a plain word.



The facets (the buttons under the chips) list, for the scans found, every kind, what they are, their anatomy, the apps for them and the years, each with how many scans have it. Choose one to add it to the search. The facets that most scans have a value for come first; a facet no scan has a value for is greyed out.

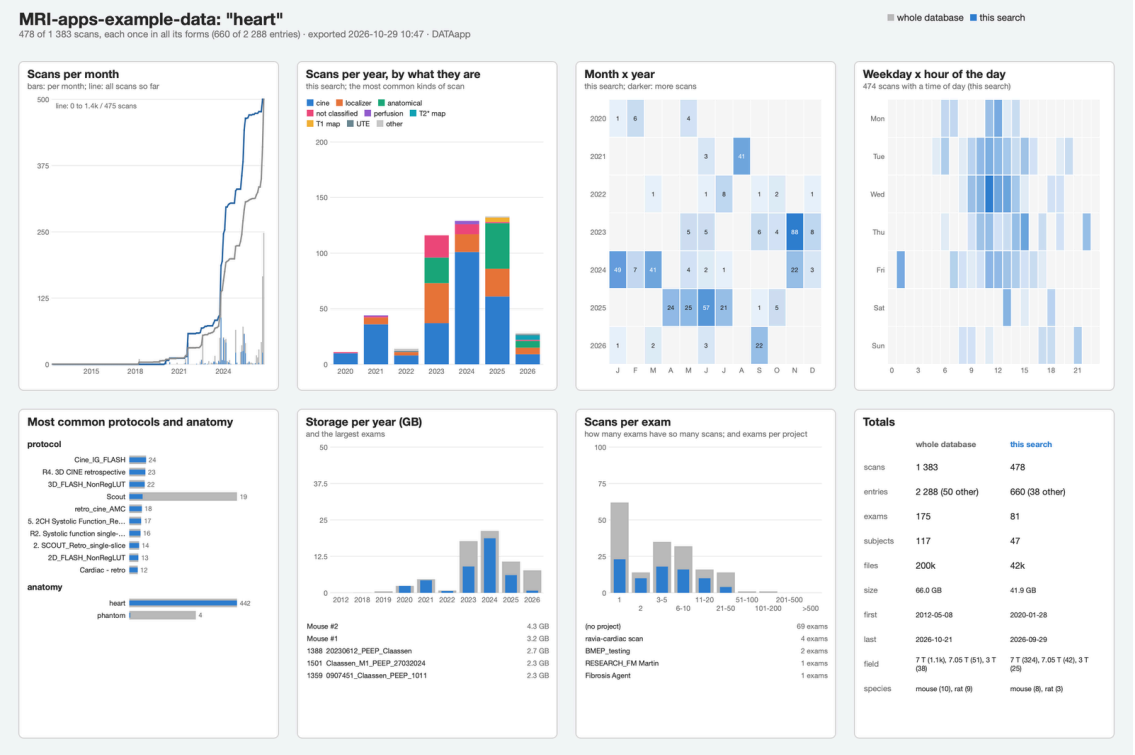
The months (the bars under the facets) show how many scans were made in each month. Point at a bar to see its month and count; click it to search that month.

The results are the exams, each with its scans under it, sorted by exam number, the highest first. An exam's line shows its number in bold with its name on a second line under it (an exam without a number shows its folder's name), how many scans were found in it and its date; a scan's line its number and name, what it is, its anatomy, its kind, its date and the apps for it. A column that almost none of the scans found have a value for is left out. Drag a column's border to make it wider. Click a column's title to sort by it; click it again to turn the order around. The arrow in the title shows the column and the direction, and DATAapp keeps them for the next start. Whatever the sort, an exam whose number you typed stays on top, and empty values go last. Sorted by exam, the scanner's numbered exams come before exams with other names, and an exam's scans stay in the order of their numbers. The line under the results says how many scans were found, in how many exams, and how long the search took. Every other exam, with its scans, lies on a faintly shaded band, so that the exams stand apart. Click an exam to open it; click a scan to choose it. Close an exam with its arrow: it stays closed while DATAapp adds new scans to the results. **Close all** (beside the line under the results) closes every exam, so that only the exams' lines show. While DATAapp adds new scans and exams, above or below, the scan you chose stays where it is in the window, and the results stay where you scrolled them. Rest the pointer on a cell to read its whole text.

Export (Dashboard) saves the scans found, with an overview of them, in a folder of its own. You choose where; DATAapp makes the folder there, named after the date, the time and your search (`DATAapp_export_2026-10-29_1432_heart`), and remembers the place for the next time. The folder holds:

- `scans.xlsx`, to open in Excel, Numbers or LibreOffice, with four sheets: **Scans**, a line per scan as the results list it (its exam, subject, number, date, kind, what it is, anatomy, protocol, sequence, matrix, TR, TE, TI, flip angle, field, files, size, species, user, project, apps, its full path on this computer, and the acquisition it belongs to); **Exams**, a line per exam; **Summary**, the numbers behind the figures; and **Search**, what you searched and when, with the totals;
- `overview.png`, a page of eight figures (below).

A scan often comes in several forms: its raw MRD file, its images, its DICOM series, a NIfTI made from it. The overview counts it once, whatever forms of it the results hold; the files and the sizes count every form.



The eight figures:

1. **Scans per month**, from the first month to the last, with a line rising to all scans so far.
2. **Scans per year**, each year's bar divided by what the scans are (cine, localizer, T2* map, ...).
3. **Month x year**: how many scans each month of each year; darker is more.
4. **Weekday x hour of the day**: when the scans are made.
5. **The most common protocols and anatomy.**
6. **Storage per year**, in GB, and the largest exams.
7. **Scans per exam**: how many exams have so many scans; and the exams per project.
8. **Totals**: scans, entries, exams, subjects, files, size, the first and last date, the field strengths and the species.

When your search lists fewer scans than the whole database, the figures show your search in blue over the whole database in grey, and the totals side by side; figures 2, 3 and 4 show your search. Without a search they show the whole database.

8. The scan chosen

The right column shows the scan chosen, in six tabs.

Summary: the scan's first image on top; under it the exam's and the scan's numbers (a scanner's image series shows the number of its scan and of its reconstruction, `23148 / 1`), the scan's name and what it is, its size (matrix, slices, frames, echoes, dynamics, coils), its timing (TR, TE, TI, flip angle, field of view, field strength), its date, its exam, its files and their size. Under it its tags as chips. When a value of the scan differs from what most scans of the same protocol have, the summary says so under **Unusual for this protocol**. At the bottom, each under a line with its heading: **Where**, the scan's folder (and the other folders switched on that hold a copy of it), with **Show in folder**, which shows it in the Finder or the Explorer; and **Open with**, why the best apps suit the scan, then a button for each app that opens it (the best first; section 9). Each button's tip gives its reason too. When there are many, the buttons take several rows and the section scrolls.

Images: the scan's images, one at a time: the slider, or the mouse wheel over the image, goes through them. **Play** runs through the images or slices, about eight a second, and starts again at the first; press it again (it reads **Stop**) to stop. A picture shows in its own colours; a GIF of several frames shows each frame as an image. **Zoom** (it stays down while it is on): drag a box over the image to zoom in on it; the zoom is kept while you go through the images of the scan. **Whole image** shows the whole image again. A raw scan shows the images the scanner made of it, when there are any. DATAapp shows the images that exist; it does not reconstruct raw data (open it with P2ROUD or the app that suits it).



Parameters: every value read from the scan's files, grouped by the file it comes from (the MRD file, the reconstruction file, the protocol, the DICOM header, ...). Type in the filter field to show only the values whose name or value holds that text. Empty and zero values are left out unless **empty and zero too** is ticked. Values that DATAapp worked out itself, such as an anatomy deduced from the exam's name, are under **deduced by the app**.



Files: the files of the scan's folder, with their size and date.

Related: the scans that belong with it: the raw data and the images of the same scan, and the other scans of its exam. Double-click one to choose it.



Notes: the notes on the scan and the corrections of what DATAapp says about it (section 10).

9. Opening a scan in an app

The **Open with** buttons in the Summary open the scan in an app at once, as if you had loaded it there yourself. DATAapp offers the apps that suit the scan, the best first; point at a button to see why it is offered:

App or program	Offered for
Retrospective	raw scans it can reconstruct: gradient echo scans with navigator samples (cine), 2D radial, 3D UTE and ZTE scans
P2ROUD	every raw scan, first for undersampled, 3D, radial, UTE and ZTE scans
MAPapp	T1, T2 and T2* mapping scans, with water & fat and QSM of multi-gradient echo scans (offered first)
T1mapp	T1 mapping scans (variable flip angle, inversion or saturation recovery)
T2mapp	T2 and T2* mapping scans (multi-echo)
ADCmapp	diffusion-weighted scans
DSCmapp	perfusion scans
SHIMapp	field maps
TRAJapp	scans made with a k-space trajectory file
ITK-SNAP, 3D Slicer	DICOM series and NIfTI files
Horos	DICOM series (macOS)
MRIcroGL	NIfTI files
MATLAB, HDFView, Excel	MATLAB, HDF5 and table files

A button of an app that is not installed on your computer is greyed out. The **Open with** list in **App settings** shows where each app and program is on this computer.

To add a program, click **Add a program ...** in **App settings**: DATAapp opens your own list of programs, `programs.json` (in the app's folder, section 13), with an example in it. Each entry gives the program's name, where it is on a Mac and on Windows, and how it is started with a scan. An entry with the name of a program DATAapp knows replaces that program's. DATAapp reads the list at its next start.

10. Notes and corrections



Notes. Type a note on the scan and click **Add note**. Everyone who searches the same folder sees it, with your name and the time.

Corrections. What DATAapp works out about a scan is a guess. When it is wrong, correct it in the Notes tab: choose what to correct (**what it is**, **anatomy**, **species**, **add a tag**, **remove a tag**, **add an app**, **remove an app**), type the right value and click **Correct**. A correction wins over what DATAapp works out, also after the scan is read again, and everyone who searches the same folder sees it.

11. Indexing

While DATAapp is open it keeps its database up to date by itself, in the background, at a low priority, so that the scanner and your work go first. It never changes your scans.

The Dashboard's **Indexing** lamps show what it does:

- **newest exams**: the newest exams of the scanner's folders, every few seconds: a new scan shows within moments.
- **changed folders**: the folders that changed, every minute.
- **all folders**: every folder, slowly, one after the other.

A lamp is green when its job is up to date, yellow and slowly pulsing while it works, orange when another computer does that job for the same folder, and red after an error (the message box says what went wrong). The check of all folders works a minute and rests a minute; while it rests its lamp is green. The bar under **Exit** says what is going on: **Newest exams ...**, **Changed folders ...** or **All folders ...** while that job works (**All folders: resting** between the check's minutes of work), and **Up to date** only when no job works and no check of all folders is under way. The message box says when a check of all folders begins and when it is done. **Pause** stops the indexing until you click **Go on**.

A file that is being written (a scan that is still running) is left until it is complete. A file of a cloud folder that is not on your computer (only online) is not downloaded: DATAapp knows it by its name only, and says so in the summary.

12. The database

DATAapp keeps what it found in a database on your own computer, so that searching is fast also when the folders are on a network drive. The Dashboard shows how many scans and exams it holds.

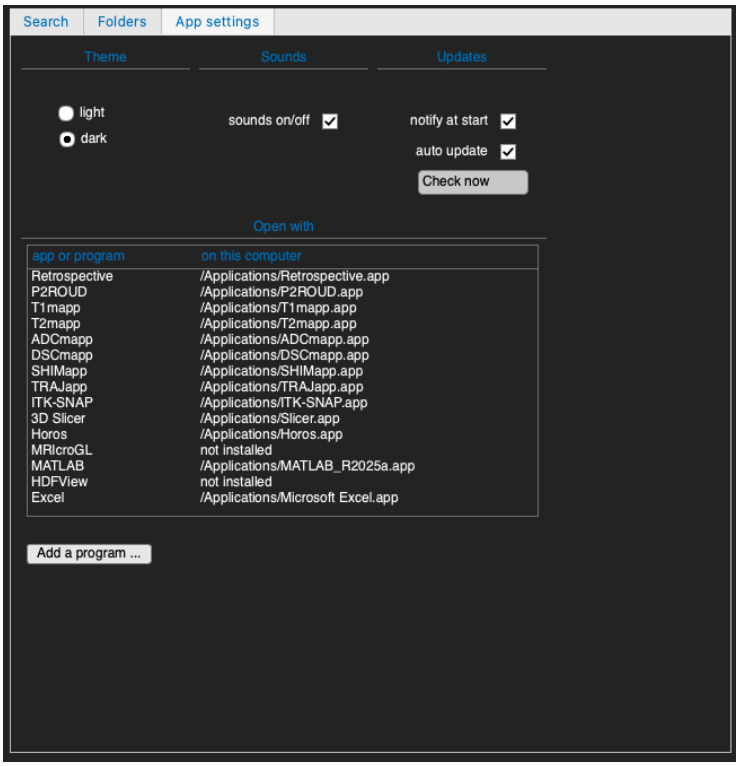
- **Backup** makes a copy of the database now. DATAapp also makes one every day while it is open, and keeps the newest of the last three days and of the two weeks before.
- **Check** checks the database for damage. A damaged database is replaced by its newest sound copy at the next start, and the message box says so. When there is no sound copy, DATAapp starts a new database and fills it again from the shared indexes and the folders.

The database and its copies are in the app's folder (section 13): macOS `~/Library/Application Support/DATAapp/database`, Windows `%APPDATA%\DATAapp\database`. Every version of DATAapp uses this one database: a new version brings it up to date, so an update leaves no old database behind. Copies that are no longer needed are removed by themselves: older copies beyond those kept, and a damaged database after a week. Your notes are also kept in the shared index of their folder and in `user_notes.json` beside the database.

13. The settings

App settings tab:

- **Theme**: **light** or **dark**.
- **Sounds**: the sounds of warnings and errors on or off.
- **Updates**:
  - **auto update**, ticked as it comes: new versions of DATAapp arrive by themselves. When one is ready, a window says so with what is new: **Install and Restart** installs it and starts the new DATAapp; **Later** installs it when you quit DATAapp, without asking again. Without an internet connection nothing is said and nothing waits. On Windows, DATAapp installed for all users can be updated only by an administrator; for other users the message box says once where to get the new version to install by hand.
  - **Check now** looks for a newer version at once, also with **auto update** off; the message box says what it found. On the Mac, **Check for Updates...** in the DATAapp menu does the same.
  - **notify at start**, ticked as it comes, counts only with **auto update** off: when a newer version is available, a window at the start offers it. **Download** gets the installer for this computer, **Skip this version** shows no window for that version again, **Later** asks again at the next start.
- **Open with**: the apps and programs a scan can be opened with, and where they are on this computer; **Add a program ...** (section 9).



The app's folder (macOS: `~/Library/Application Support/DATAapp`; Windows: `%APPDATA%\DATAapp`) holds:

- `0.2.1p/dataappSettings.json`, the settings file: your theme, sounds, **auto update**, **notify at start**, your last search and the widths of the result columns, for the next start; a new version takes them over, and the message box says so. A settings file that cannot be read is kept beside it under another name, and the app starts with the standard values and a warning. It also holds `dataLocations`, the folders added by themselves when they exist (set to the lab's network share, `L:\Basic\divh\BMEPH\Gustav` on Windows and its address `smb://amc.intra/opslag/basic/divh/BMEPH/Gustav` on macOS, found wherever the Mac has it connected; add your own, or empty the list), and `dataLocationsAdded`, those added so far.
- `0.2.1p/dataapp-log-<date-time>.txt`, a log of each session with every message the message box showed and its time; the newest 100 are kept.
- `programs.json`, your own programs for **Open with** (section 9), once you have added one; every version reads it.
- `database/`, the database and its copies (section 12).

New versions: at its start DATAapp looks on GitHub for a newer version and says in the message box which version this is and where a newer one is. In the settings file, `checkForUpdates` (`false`) turns this off and `versionCheckTimeout` gives the seconds it may take. Change them in the file while the app is closed.

14. Credits

DATAapp uses Python, SQLite, NumPy, SciPy, PySide6 (Qt), pyqtgraph, pydicom, NiBabel, h5py and Pillow. Its window is built as those of the other MRI apps of Amsterdam UMC (Retrospective, P2ROUD, T1mapp, T2mapp, ADCmapp, DSCmapp, SHIMapp, TRAJapp). The image in its icon is a T2-weighted MR image of a mouse brain made on the MR Solutions scanner of Amsterdam UMC.